SELECTED WORK / AI TEAMS

Boundaries that can be inspected

Lux Eligarden | AI-assisted systems design | Private review edition | 26 October 2026

A focused portfolio in problem formulation, architecture, governance and evaluation design. Prepared for employment and scoped collaboration discussions; no independent programming proficiency or research-validity claim is inferred.

Work sample	Assessable capability	Evidence boundary
Aegis 0.4.2 / flagship	Verify meaning as well as bytes. Separate modeled conformance from real authority.	Code and historical reports; overall validation blocked.
Orchestrator 0.4.1	Separate decision, permission, reservation and completion.	Local implementation; acceptance incomplete.
Mirror AI 1.3.0	Keep observations, alternatives and unknowns distinct.	Authored synthetic graphs; no empirical accuracy claim.

Why these selections

Each sample has a concrete mechanism, inspectable source records and explicit limits. Selection did not depend on grand scope or high version numbers. Broader personal-model, companion and civilizational architectures would need more context and authority reconciliation.

How to read the evidence

Historical test results are labeled as such. Diagrams and state sequences are editorial explanations. No source runtime tests, user studies or benchmarks were newly executed for this portfolio. The accompanying repository includes selected source files, commit/hash provenance and a fuller case study.

READY FOR OWNER REVIEW. External sharing remains a separate owner decision. No public release, deployment or collaborator invitation is part of this preparation.

FLAGSHIP / AEGIS

Verifying meaning as well as bytes

Aegis 0.4.2 | Bounded constitutional-AI research and education laboratory

The concrete problem

A changed decision record can be internally hash-consistent while misrepresenting what the registered workflow would produce. A second failure is treating a valid modeled artifact as a legitimate real-world authorization. Aegis addresses the narrower problem of reconstructible, bounded counterfactual branches over an immutable episode.

Users, context and decomposition

A learner or reviewer inspects CEE-001 and constrained alternatives using a Python/FastAPI reference with static Gate, Grow and Map surfaces. The package is not connected to live governance or execution runtimes. Intended educational use is not evidence of adoption or learning benefit.

Boundary	Checks	Cannot prove
Integrity	Hashes, references, artifact identity and root consistency.	That a consistent story is the correct story.
Semantic reconstruction	Agreement with the registered branch engine and source parent.	That the engine models reality accurately.
Exact authority content	Bindings to exact engine, registries, template, invariants and policies.	Institutional legitimacy or independent producer trust.

Design trade-off

Hash-only verification is simpler but insufficient for the source's tampering cases. Exact content binding improves traceability but increases migration burden. Bounded templates improve inspectability while constraining exploration. A live governance service would need evidence and authorities this package does not establish.

Sources: sources/aegis/README.md, RELEASE_MANIFEST.json, docs/IMPLEMENTATION_STATUS.md and docs/CONTENT_ADDRESSED_SEMANTIC_AUTHORITY.md.

FLAGSHIP / WORKED WALKTHROUGH

A denial is not an execution

Read-only annotation of existing fixtures and historical source reports; not a new run.

1. **Choose the existing template**
   lab-004-reviewer-denies over the preserved CEE-001 episode.
2. **Apply the approved mutation**
   governance.review_outcome = deny. No denial rationale is invented.
3. **Inspect the stated causal effects**
   Authorization absent; permit/execution blocked; completion withheld.
4. **Compare verification layers**
   Stored branch conformance passes against exact authority content.
5. **Inspect the separate forgery record**
   Nine historical variants: integrity passes; semantics/composite fail.

What this example demonstrates

Internal consistency and semantic conformance are distinct checks. The stored report supports a bounded claim about the registered deterministic engine. It does not demonstrate empirical truth, institutional legitimacy, real authorization or live execution.

What is deliberately missing

The template explicitly leaves the hypothetical denial rationale unspecified. This portfolio does not create one. The forgery records are not an independent security audit or a complete attack-surface evaluation. The walkthrough substitutes for an unavailable runnable demonstration/video in this curated packet.

Sources: data/lab_templates/lab-004-reviewer-denies.json; reports/branch-semantic-conformance-report.json; reports/semantic-forgery-test-report.json. All are included under sources/aegis/.

FLAGSHIP / EVIDENCE AND EVALUATION

Passing checks can coexist with blocked readiness

Keep architectural scope, implementation evidence and validation status separate.

Claim axis	What the selected source supports
Architecture	A bounded laboratory candidate with explicit operating modes and no real authorization/execution.
Implementation	Existing source, templates, verifiers, static frontend and tests.
Historical tests	Validation report records 433 passing tests; technical checks passed.
Overall validation	Blocked. Full-pipeline, browser/accessibility, container and supply-chain gates remain unresolved.
Readiness	Public-pilot deployability false; production readiness false; institutional legitimacy not verified.
Empirical validity	No educational-benefit study, real governance outcome study or independent security certification established.

Important source conflict

The title includes "Verified Public-Pilot," but the manifest and validation report leave deployability false. This portfolio follows those narrower flags, preserves the discrepancy and makes no new global version ratification.

Proposed evaluation - no results

Independently rerun existing positive and negative fixtures at fixed versions. Compare integrity-only and composite classifications, recording unexpected acceptance, rejection, unavailable checks and reconstruction differences. A later consenting-reviewer study could assess whether people distinguish model conformance from legitimate authority; no study was performed here.

Sources: sources/aegis/RELEASE_MANIFEST.json; reports/validation-report.json; docs/CONTRADICTIONS_AND_DECISIONS.md. Proposed protocol: EVALUATION_PLAN.md.

SUPPORTING EXAMPLES

Authority and interpretation boundaries

Two distinct mechanisms; no new integration between the selected systems is claimed.

Orchestrator 0.4.1 | A permit is not completion

A native promotion permit binds exact candidate content, parameters, expected current version, operation, executor and handler. Its source profile separates atomic lease reservation from completion evidence and rejects replay after completion. Durable receipt and execution identity matter; local signing is not production trust infrastructure.

Historical evidence: 173 Orchestrator tests passed. Line coverage 77.79% and branch coverage 61.92% remain below its declared 94% / 90% gates. Per-surface and cross-process fault matrices are incomplete. Acceptance, pilot and production status remain incomplete/false. No benchmark was rerun for this packet.

Mirror AI 1.3.0 | A hypothesis remains a hypothesis

An included authored synthetic graph labels an unexpected sword metaphor as an observation and "Prompt carryover" as a candidate explanation. Empirical claims and calibrated probability are false. The service validates source-linked graph data; it does not discover the underlying mechanism or infer a person's cognition.

Historical evidence: 108 successor tests passed after a test-only correction to an expired inherited fixture. Real Gaia compatibility, production trust and empirical interpretation accuracy remain unresolved. The release did not change an external canonical pointer.

Internal critique

Explicit state and interpretation boundaries improve inspectability but add lifecycle and annotation costs. Faithfully reproducing a flawed engine or clearly rendering an unsupported hypothesis does not make either true. This is an AI-assisted internal review, not external validation.

Sources: sources/orchestrator/IMPLEMENTATION_STATUS.json and CONFORMANCE_REPORT.json; sources/mirror/docs/IMPLEMENTATION_STATUS.md, evidence/validation_report.json and canonical/contradiction_ledger.json.

CONTRIBUTION / PROVENANCE / REVIEW

What this work sample can support

Design direction and AI-assisted development are described separately.

Contribution statement

The corpus supports Lux's role in concept direction, requirements, constraint-setting, selection/rejection of alternatives and iterative refinement. AI assisted formalization, documentation, code and checking. This portfolio does not establish that Lux independently wrote or debugged the implementation. Exact individual authorship of each code change has not been independently established.

Source checkpoints

Aegis-Constitutional-AI-Lab: 1ff5267fb367
ArchEngine-Orchestrator: f00e122be68f
Mirror-Systems-Archive: 1bbfc4987299
Full commit identifiers, exact package members and SHA-256 values are included in SOURCE_MANIFEST.json. Case studies and selected source snapshots can be read without access to the preservation repositories.

Rights and sharing

No new open-source or commercial-use grant is created. Existing component notices and prior licenses retain their scope. No claim of exclusive ownership of abstract ideas or guaranteed economic return is made. This curation is not a legal clearance audit.

Ready now / optional later

Ready now: a private review portfolio with flagship, two supports, mechanism walkthrough, source evidence, proposed evaluation and internal critique. Optional later: owner corrections to the contribution wording, existing external evidence if available, or recipient-specific tailoring. No additional material was available now, and no unavailable item blocks this review copy.

READY FOR OWNER REVIEW, not owner-approved for external sharing. Current framing: employment and collaboration; AI and game systems design. Source systems and source repositories were not extended.